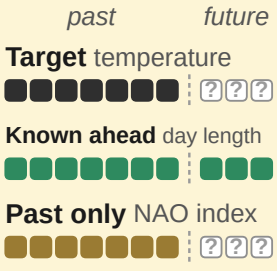
How Chronos-2 makes a forecast

Amazon · 120M parameters · encoder-only transformer · zero-shot

1. One row per series

The target and its covariates form one group. A ? is a value not known yet.

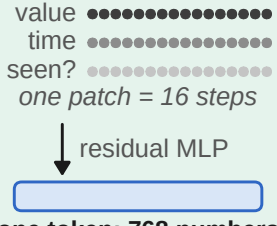


Several targets, such as five metal prices, can share one group.

2. Scale, then cut into patches

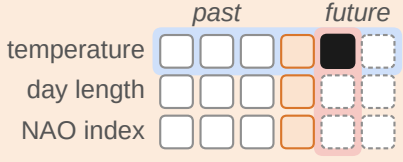
Each row is scaled by its own past mean μ and standard deviation σ, then squashed by arcsinh.

$$\tilde{x} = \operatorname{arcsinh}\left(\frac{x-\mu}{\sigma}\right)$$



3. Twelve blocks mix the tokens

Each layer starts with RMSNorm and has a shortcut around it. The orange REG token marks where the future starts.



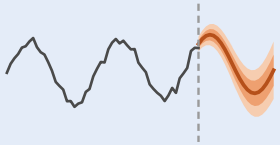
Time attention: along the row, all past and future patches (RoPE positions)

Group attention: down the column, the other rows at the same patch

Feed forward: each token on its own, 768 → 3,072 → 768, ReLU

4. All quantiles in one pass

After a final RMSNorm, an output network turns each future token into **21 quantiles** for each of its 16 steps.



$x = \mu + \sigma \cdot \sinh(\tilde{x})$ restores the units; the median is the point forecast.

Group attention lets covariates and related series shape the forecast.

12 blocks · width 768 · 12 heads · context up to 8,192 steps · up to 1,024 steps per pass

In the notebooks: last 2,048 days or full monthly history · one group per forecast · quantiles 0.1 to 0.9 kept